

DOC-03

Data Relationship Diagram

Module-level data flow and business logic connections



Project	GPA-ERP V5.0
Type	Construction Cost Control ERP
Stack	FastAPI · Next.js 14 · PostgreSQL
Version	V5.0

GPA-ERP — Data Relationship Diagram

This diagram shows how **data flows between modules** — not just foreign key links, but the business logic that connects them.

High-Level Module Flow

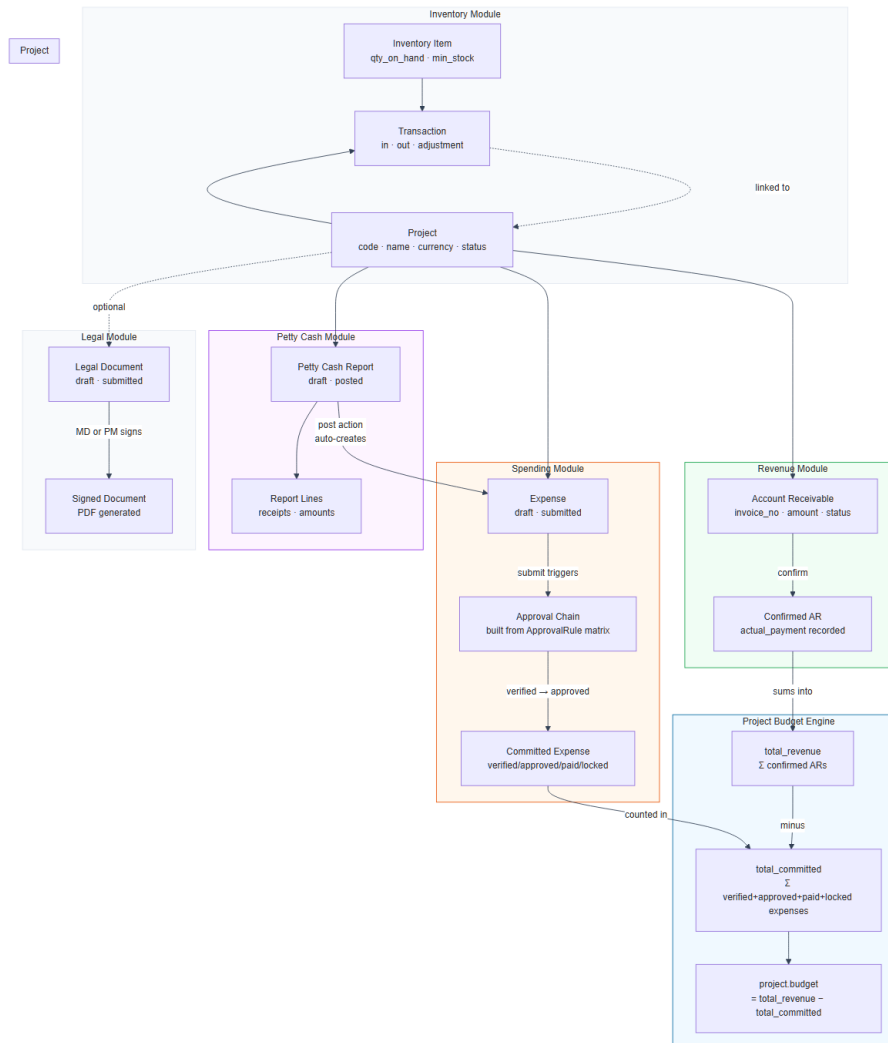


Figure 1

Budget Calculation Detail

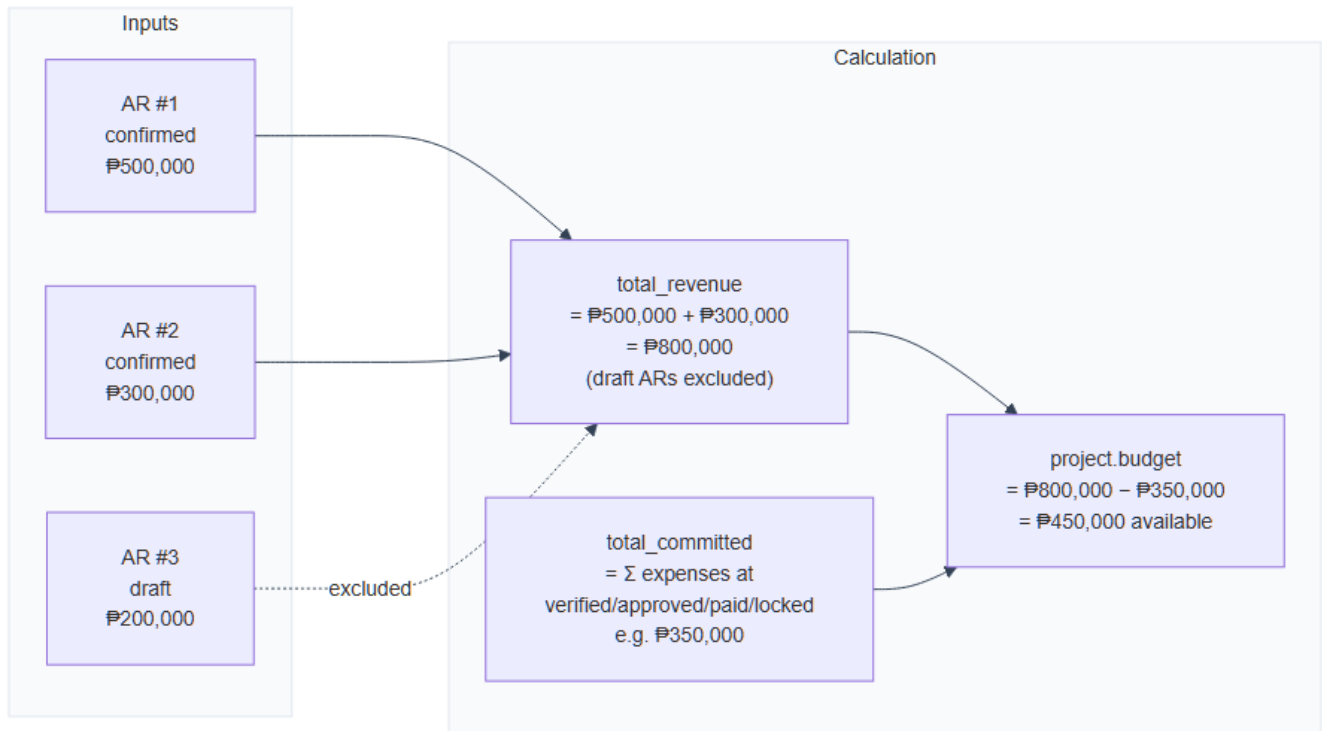


Figure 2

Key rule: Only **confirmed** ARs count toward revenue. Only expenses at verified, approved, paid, or hard_locked count toward committed spend. Draft/submitted/rejected expenses do NOT reduce the budget.

Expense Approval Chain Resolution

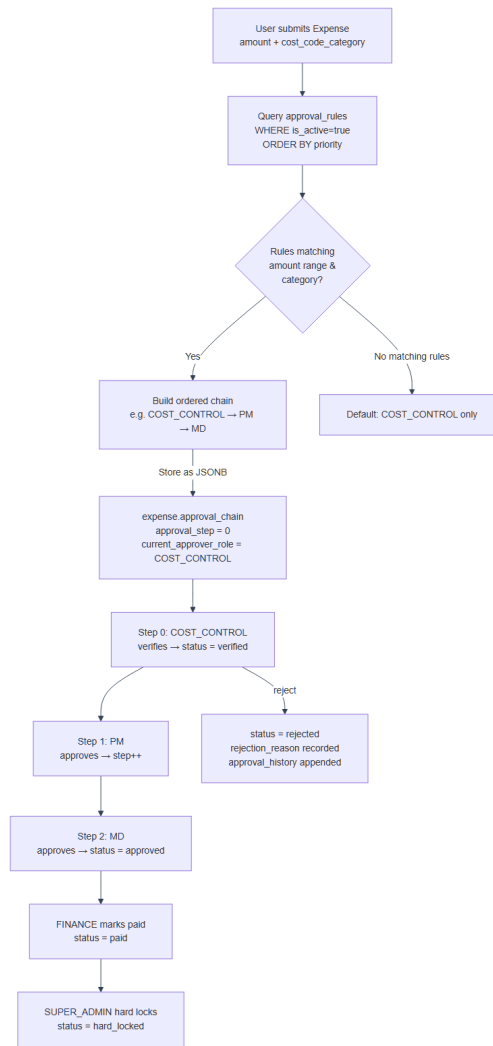


Figure 3

Petty Cash → Expense Pipeline

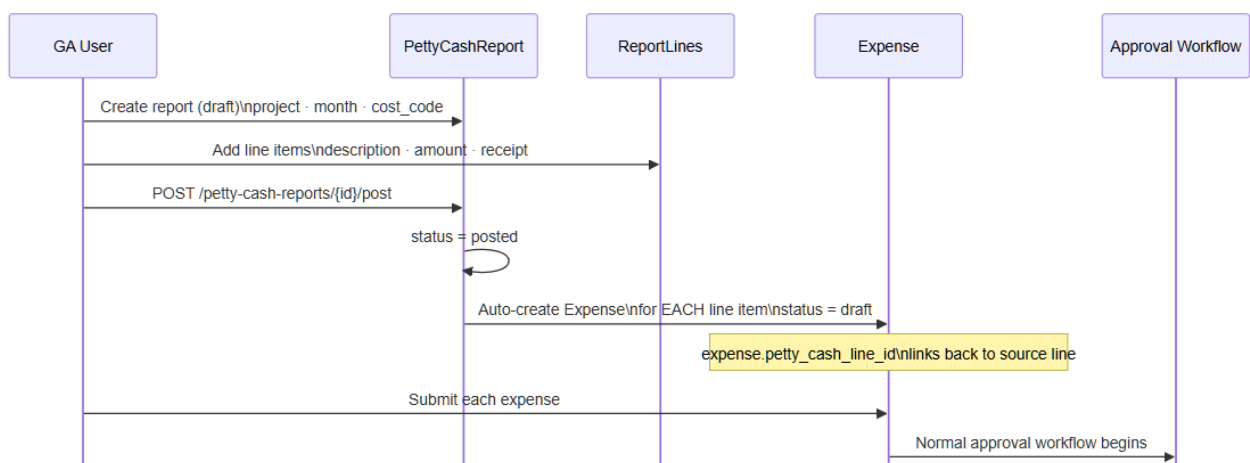


Figure 4

Legal Document Lifecycle

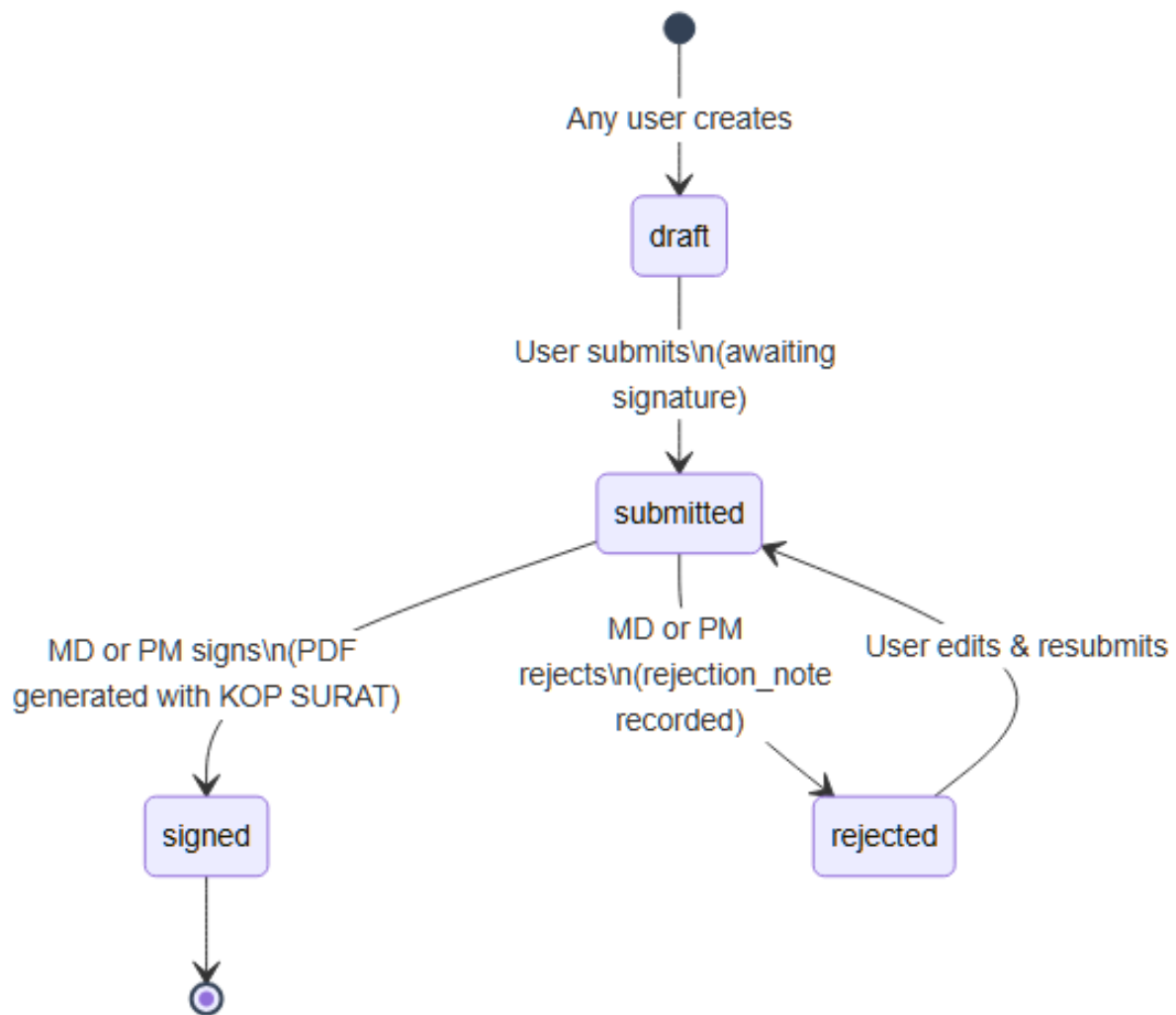


Figure 5

Inventory ↔ Project Link

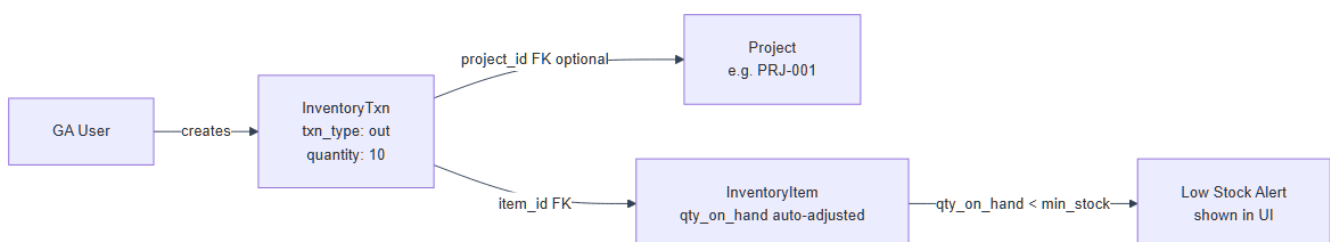


Figure 6

Inventory transactions are **project-linkable but not project-required**. Items consumed on-site can reference a project; warehouse adjustments typically don't.

Audit Trail Flow

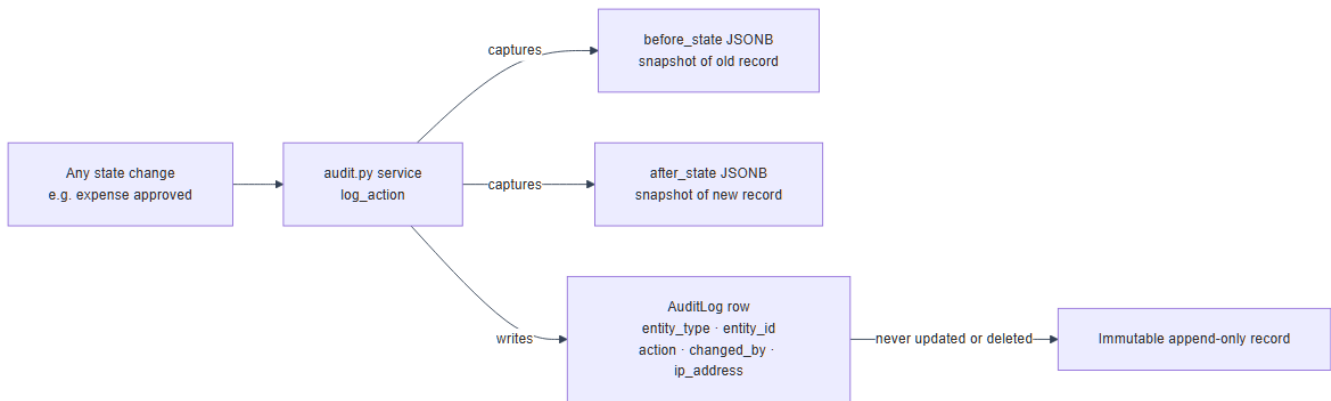


Figure 7

The audit log is exposed read-only at GET /vault/audit-log and is only accessible to SUPER_ADMIN and COST_CONTROL.

Cross-Module Data Summary

Source	Feeds Into	How
account_receivables (confirmed)	project.total_revenue	Hybrid property sum
expenses (verified/approved/paid/locked)	project.total_committed	Hybrid property sum
project.total_revenue – total_committed	project.budget	Computed on every read
petty_cash_report_lines (when report posted)	expenses	Auto-created, one per line
approval_rules (at submit time)	expense.approval_chain	Resolved once, stored immutably
inventory_txns	inventory_items.qty_on_hand	Applied by router at transaction create
Any state change	audit_logs	Via audit.py service